

بسم الله الرحمن الرحيم

Intestinal Protozoa

I - Entamoeba Histolytica

First Course
(Lecture 3)

Entamoeba histolytica

Objective:

- **Disease:** Amoebic dysentery (intestinal & extra intestinal amoebiasis).
- **Causative agent:** *Entamoeba histolytica*
- **Infective stage:** mature cyst stage from patient's stool.
- **Route of entry: Orally**
- **Habitat:** large intestine, mainly in the caecum and colon.
- **Host:** only human body.
- **Diagnostic stage:** Cyst and trophozoite stage in patient's stool.
- **Source of infection:** only human from patient's stool.
- **Movement:** is by pseudopodia.
- **Reproduction:** a sexually by binary fission



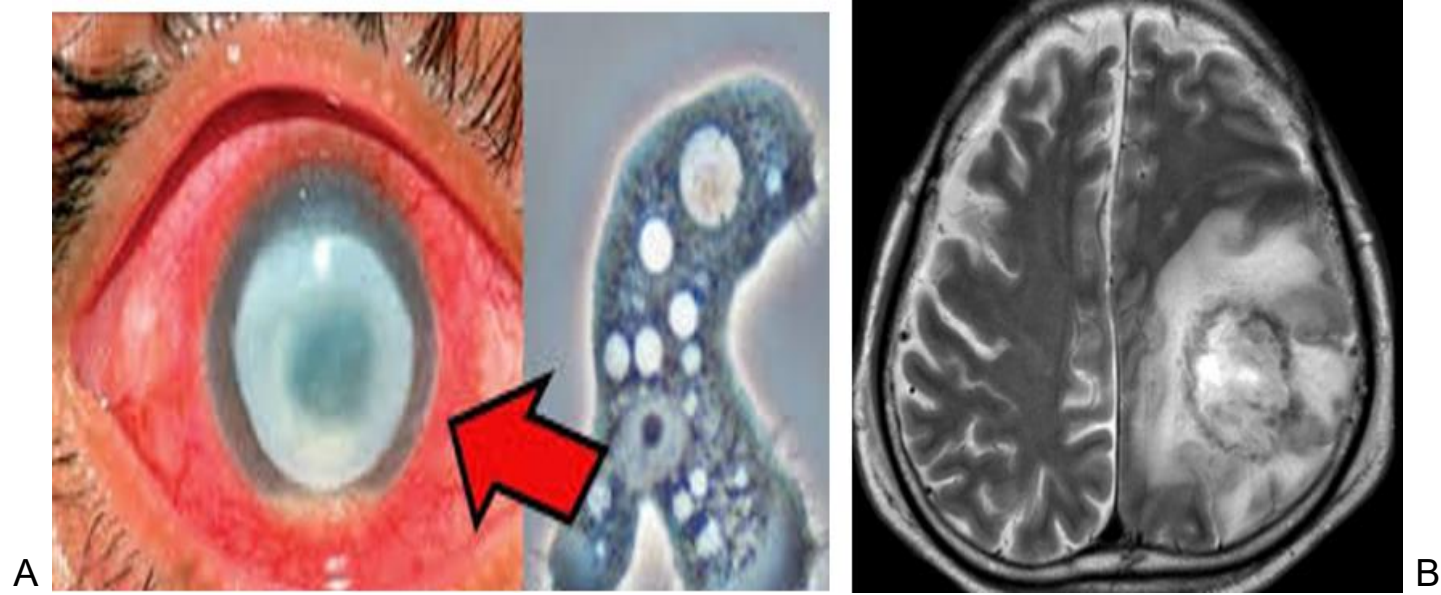
Entamoeba histolytica, causing intestinal (dysenteric) amoebiasis



Introduction

Entamoeba histolytica causes **amoebic dysentery** (intestinal & extra intestinal amoebiasis), was discovered by Losch in 1875, and in 1890 William Osler reported the case of a young human with dysenteric feces, who later died of liver abscess, it is the third leading cause of mortality after **malaria** and ***schistosoma***

All intestinal amoeba are non-pathogenic except for ***entamoeba histoloytica*** which causes **amoebic dysentery**, and ***acanthamoeba*** which causes **keratitis** and **granulomata encephalitis**.



Acanthamoeba, causes A. keratitis (trophozoite stage in eye), B. Granulomata encephalitis.

Structure and reproduction

Entamoeba histolytica are structurally simple protozoans which have no fixed shape, irregular, and active motile, move in one direction by pseudopodia formed from ectoplasma, it's employed for locomotion and engulfment of food by phagocytosis.

Cytoplasm of ***entamoeba histolytica*** include outer layer called ectoplasma which is clear, and inner layer called endoplasma, it is granular and contain one true nucleus and food vacuoles. Nucleus is spherical, (4-6 μm) in size and contains central karyosome, and surrounded by nuclear membrane and reproduction occurs by binary fission.



Entamoeba histolytica have two important stages when developed in human life cycle (trophozoite and cyst stage).

Precystic stage: trophozoite undergo encystement in intestinal cavity.

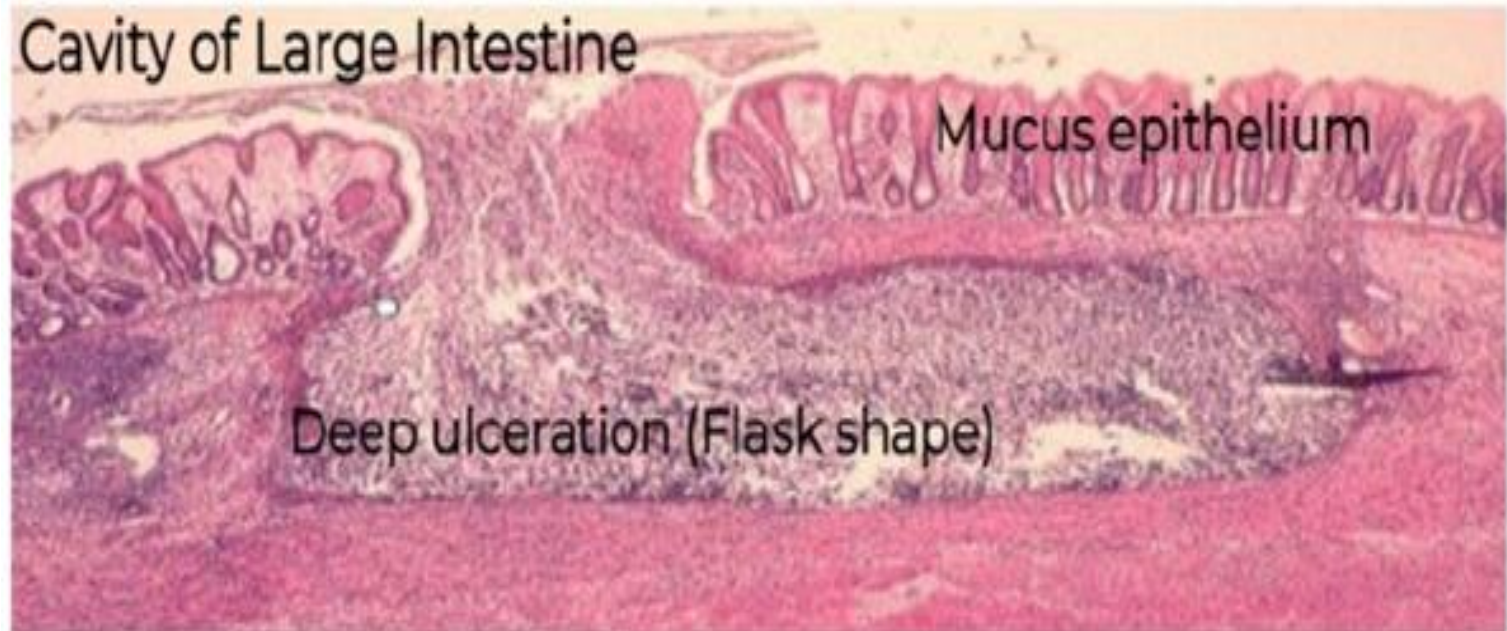
Cyst stage: trophozoite surrounded by specific membrane.

Metacystic stage: is excystation of cyst (rupture or damaged) by function of alkaline and trypsin in the cavity of intestine and liberate 4 nuclei are surrounded by cytoplasm, and form 4 small metacystic trophozoite.



If exystation take place in small intestine, the metacytic trophozoite does not colonize (infection) there but are carried to large intestine in the caecum and causes cutaneous (muco) ulceration which develop to deep ulceration called flask-shaped ulceration which extend deeper than submucosal and muscular layer, but do not extend to subserosa of the intestinal layer.

Trophozoite of ***entamoeba histolytica*** spread laterally and extend into the surrounding healthy tissue in submucosa causing extensive undermining and patchy mucosal loss, the ulcer may be involved the muscular of colon causing perforation and peritonitis, blood vessel erosion may cause hemorrhage, flask shape ulceration have mouth and neck being narrow and base layer been rounded.



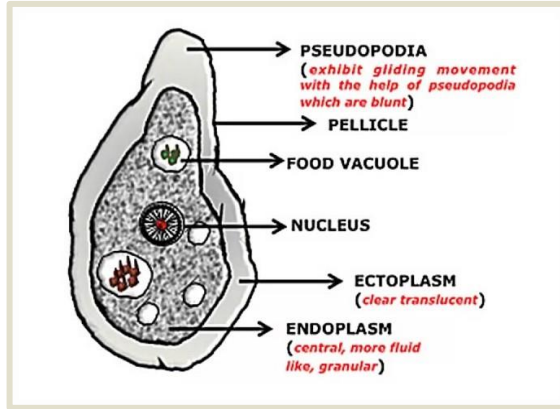
Flask shaped ulceration



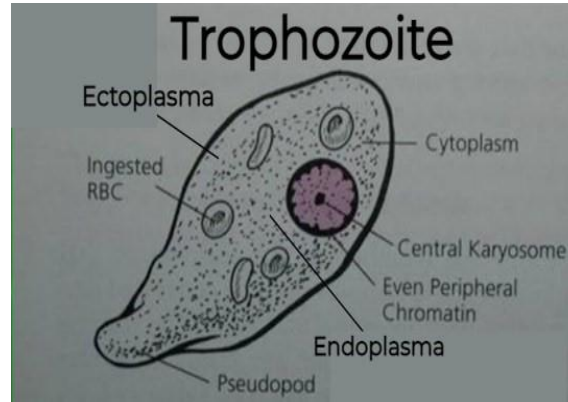
The ectoplasma function in locomotion, excretion of wastes and defense mechanisms, while the endoplasma is concerned with metabolism.

It contains food vacuoles, and excretory vacuoles (collect waste products and discharge them exterior cell by bursting through the ectoplasm), stored food are in the form of carbohydrates or protein.

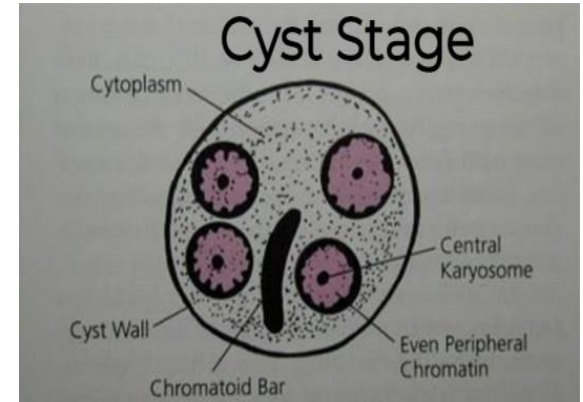
A



B



C



A. Morphology (structure) of *E. histolytica*, B. Cyst stage, C. Trophozoite stage



Life cycle

Entamoeba histolytica need one host (human body) to complete the life cycle. Humans acquire infection when swallowing or ingesting food and water contaminated with mature cyst as infective stage from stool of patients or carriers, if trophozoite of ***E. Histolytica*** are ingested they will be killed in stomach by action of gastric juices, while the infective stage (cyst) reach to stomach, the cyst wall are resistant to the acidity or gastric juice of the stomach.



The cysts pass through the stomach undamaged and enter the cavity of small intestine, some of the cysts get damaged (ruptured) and the excystation occurs, releasing four nucleated amoeba and multiply immediately, without infecting the wall of small intestine but migrate to cavity of large intestine.

the cyst wall get damaged due to the alkaline condition and the action of trypsin leading to excystation of cysts and liberation or release of four nucleated amoeba from cyst of ***Entamoeba histolytica***, each one gets surrounded its own cytoplasm to form four small trophozoites and immediately grow and multiply by binary division to become eight small trophozoites called metacytic trophozoites which start growing and developing in large intestine.



All metacytic trophozoites starts invading (penetrate) the columnar epithelial cells in the wall of lower part of large intestine in mucosa and submucosa tissue mainly in the wall of caecum and colon to form muco-ulceration and developed to deep ulceration (flask shape ulceration) in tissue wall by the aid of protolytic toxins secreted from the trophozoites. The flask shaped ulceration are resemble a bee hive, the number of trophozites increases in deep ulceration with pus cell and RBCs.



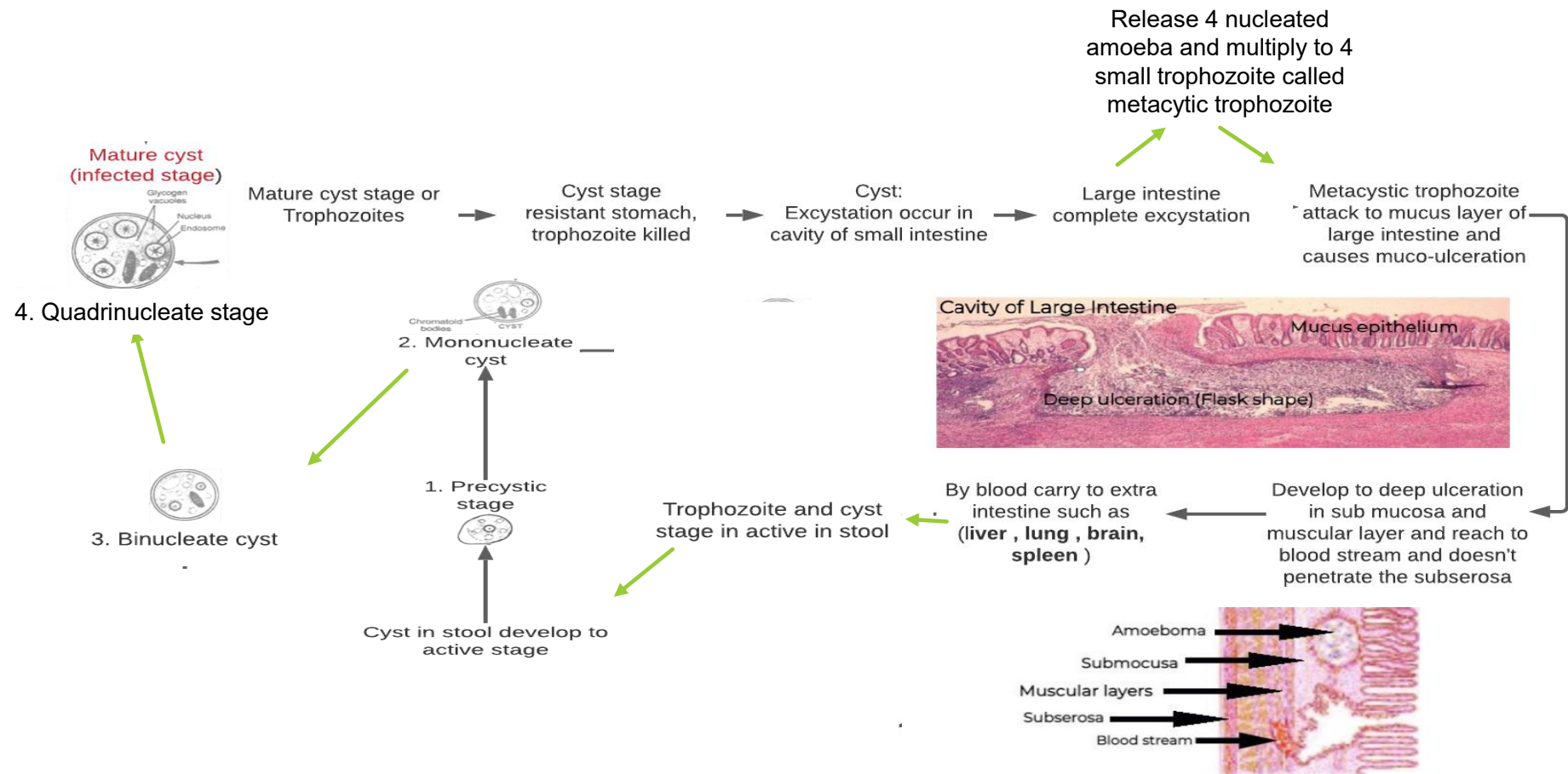
And the trophozoites are carried with bloodstream in submucosa and muscular layer to other extra intestinal sites where they reach to different organs of humans such liver, lungs, brain, spleen and etc.

The penetration of trophozoites in the columnar epithelial cells in colon is facilitated by the motility of trophozoites and the tissue lytic enzyme (histolysin) which damages the mucosal epithelium, the factors affecting virulence of trophozoite like amoebic cysteine and lectin and the host factors like malnutrition, alcoholism and immunodeficiency.

The trophozoites do not extend deeper than submucosal and muscular layer of large intestine

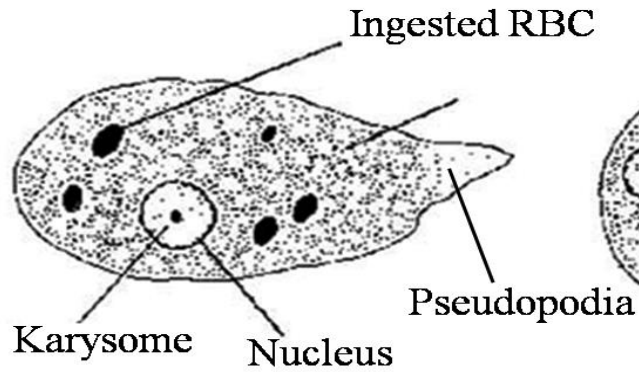


Some of trophozoites in the cavity of large intestine are passed with stool outside the body, and another trophozoite under unsuitable condition develop into precytic forms to become cyst stage (inactive stage) which are passed with stool outside the body, under suitable conditions in soil, they convert to active cyst stage, the inactivated stage has uninucleus, then converted into binucleated cysts with two nucleus and they are rapidly converted to mature cyst (active) stage with four nucleus called quadrinucleate (infective stage).

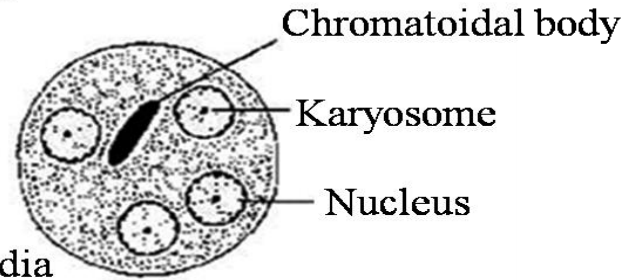


Schematic diagram of life cycle for *E. histolytica* in human host and in soil

A- Trophozoite



B- Cyst



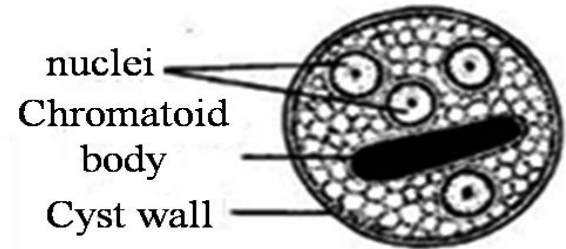
C- Development of cyst



A. Uninucleate

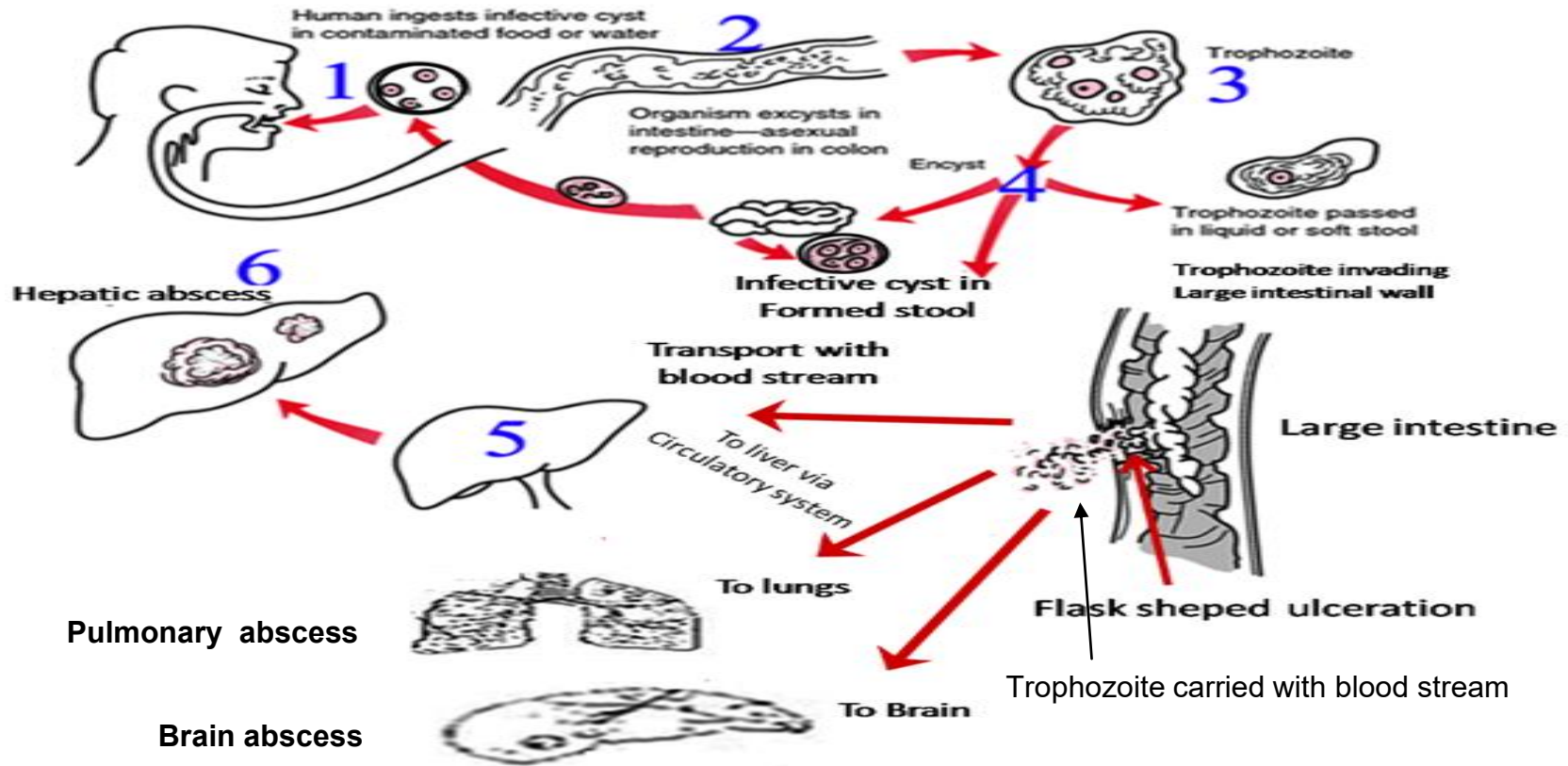


B. Binucleate



C. Quadinucleate

Schematic diagram of amoeba stages.



Schematic diagram of development cycle of *E. histolytica* within the host.

Pathogenesis and clinical features

Entamoeba histolytica causes **intestinal (dysenteric) amoebiasis** and **extra intestinal amoebiasis**. Incubation period of ***E. histolytica*** ranges between several weeks to several months. 1-4 months amoebiasis can present in different forms and degree of severity, depending on organ affected and the extent of damage caused. In **intestinal dysenteric amoebiasis**, the clinical symptoms appear when the trophozoites invade the mucosa and submucosal tissue of wall of large intestine (colon), and causes two types of ulceration, Small muco-ulceration is superficial ulcers involving only mucosa, and deep ulceration (flask shaped ulceration). The mucosa between the flask shaped ulceration is usually normal.



Some of cyst stage or trophozoite of *Entamoeba histolytica* remains as a commensal in the cavity of intestine (small or large intestine) without causing any infection of the wall, and the persons become carriers or reservoir host and called **non-dysenteric amoebiasis**.

A localized granuloma is called amoeboma is presented as palpable mass in the wall of rectum. The amoeboma or amoebic granuloma may be mistaken for are malignant tumor.

The clinical classification of amoebiasis are characterized by asymptomatic colonization of *Entamoeba histolytica* without invasion in mucosa and submucosa of colon and Symptomatic infection include invasion with mild symptoms in intestine which is the primary infection



Dysenteric (intestinal) amoebiasis

When *Entamoeba histolytica* trophozoites form deep ulceration that occurs most commonly in the wall of the caecum and colon with necrotic lesions, with the trophozoites at the periphery of the anus and causes amoebic colitis.

The clinical symptoms like abdominal cramps, abdominal pain which is mostly in the lower quadrant of the abdomen especially in the right side, without high fever, severe hemorrhage and mild intermittent diarrhea with blood (bloody stool) with pus cells RBC and offensive odor. Period of diarrhea alternate with constipation are common causes of **amoebic colitis**. **Dysenteric (intestinal) amoebiasis** associated with bacillary dysentery, but with high fever without blood.



When the trophozoites reach the bloodstream in the submucosa, occasionally the ulcers may involve the muscular and serous coats of the colon causing **perforation** and **peritonitis, blood vessel erosion** and may cause **hemorrhage**.

Blood vessels are characterized by abscesses in different human organs like liver, lung, kidney, brain, spleen, and adrenals spread to brain leads to severe distraction of brain tissue and is fatal.

Clinical Feature	Amoebic dysentery	Bacillary dysentery
Clinical on host	Slow & localized	Acute & generalized
Fever	Absent	Present
Toxicity	Absent	Present
Tenesmus	Present	Absent
Stool frequency & nature	6-8 /day with blood & mucus not adherent to container	Over 10 /day no or little blood & mucus adherent to container
Odor	Offensive	Nil
Color	Dark red with RBC	Bright red
Reaction	Acid	Alkaline
Microscopy motile bacteria	Absent	Present
Motile amoeba	Present motile trophozoite with ingested RBC	Absent
Charcot-leyden crystals	Present	Absent

Differential feature between amoebic dysentery and bacillary dysentery



The complication of intestinal amoebiasis are:

1. **Fulminant amoebic colitis:** toxic mega colon, perianal ulceration and amoeboma
2. **Extra intestinal amoebiasis:** it is complication of intestinal amoebiasis called metastatic amoebiasis spread by hematogenous through lymphatic and causes abscesses amoebiasis in different human organs.



The most important types of extra-intestinal amoebiasis includes:

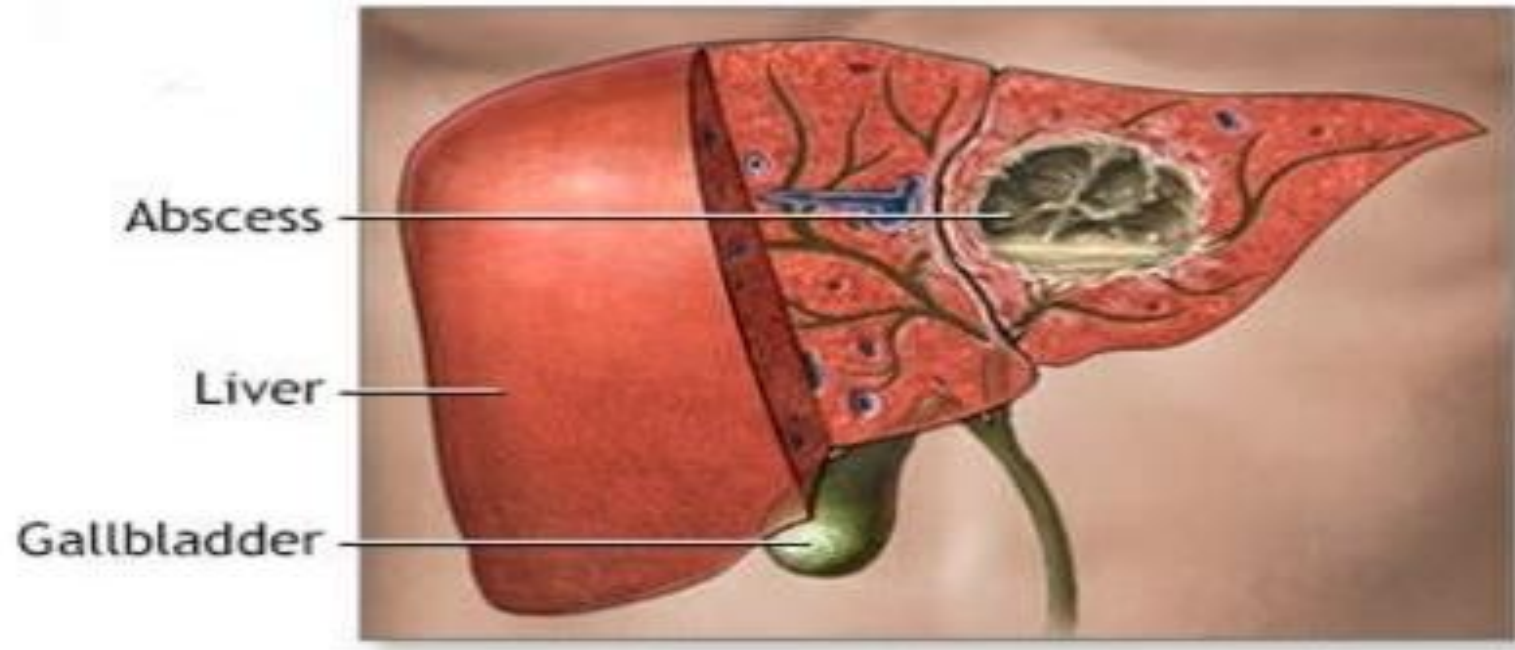
I. Ameobic liver abscess: it is the most common extra intestinal complication of *Entamoeba histolytica*, this often occurs without a history of recent diarrhea, the abscess is usually found in the right hepatic lobe more often than the left lobe.

The clinical features include pain of the right lobe of liver (epigastric pain), malaise, fever, high sweats, anorexia, weight loss and pain in the right shoulder. The aspiration from liver gives chocolate brown color because of the mixture of blood and necrotic liver tissues. Several patients with amoebic colitis develop an enlarged tender liver without detectable important of liver function or fever.



Liver amoebiasis it is acute hepatic involvement, may be due to repeated invasion by *Entamoeba histolytica* from an active colonic infection or to toxic substances from the colon reaching the liver, it is probable that liver damage may not be caused directly by the *Entamoeba histolytica*, but by lysosomal enzymes and cytokines from the inflammatory cells surrounding the trophozoite.

Untreated **ameobic liver abscess** tend to rupture into adjacent tissue through the diaphragm into the lung or pleural cavity, pericardium, peritoneal cavity, stomach intestine, inferior vena cava, or externally through abdominal wall and skin.



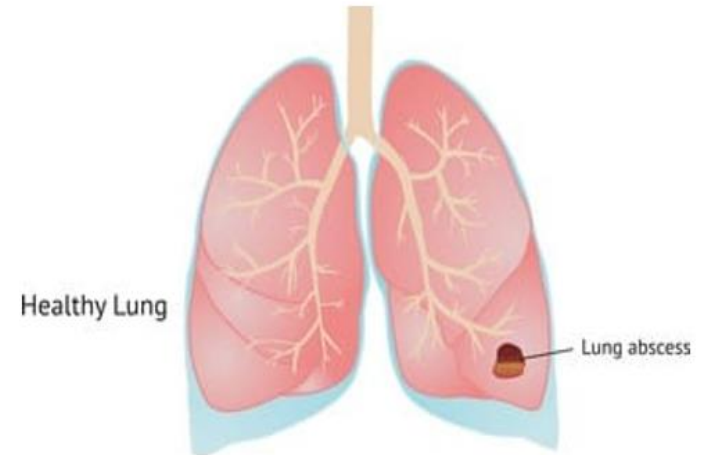
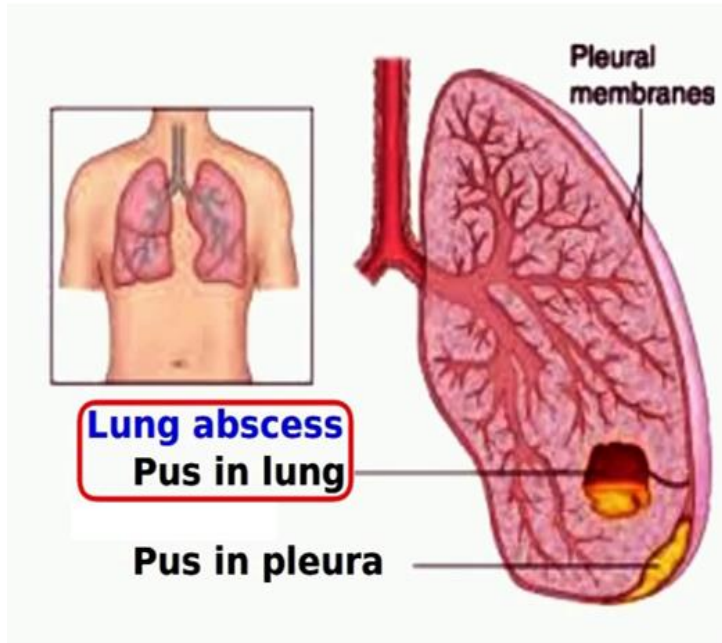
Amoebic liver abscess



2. Amoebic lung (Pulmonary) abscess: This is usually secondary to amoebic liver abscess. It occurs usually as a consequence of rupture of hepatic abscess amoebiasis into the chest cavity.

A large amount of hepatic trophozoites may penetrate the right side of the diaphragm and rupture into the chest cavity and the lungs then into the bronchus. The abscess passing through the broncho-hepatic fistula.

The clinical diagnosis of amoebic lung (Pulmonary) abscess are fever, severe pleuritic, chest pain, dyspnea, non - productive cough, and chocolate brown sputum.

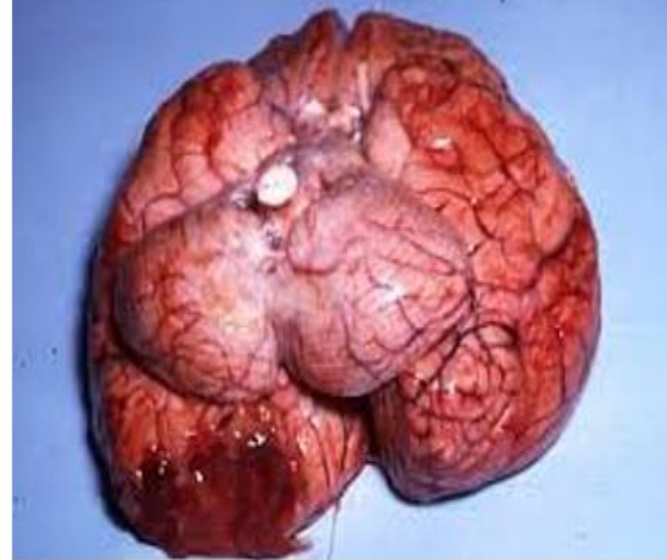
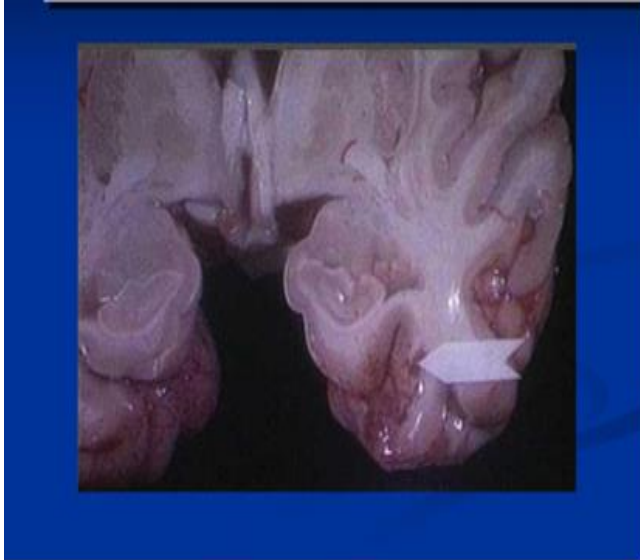


Amoebic lung (Pulmonary) abscess



3. Amoebic brain (cerebral) abscess: metastatic amoebiasis involvement of distant organs is by hematogenous spread and through lymphatics abscesses in kidney, brain, spleen, and adrenals have been noticed.

Spread to brain leads to severe destruction of brain and is fatal. This is usually due to a single lesion located in the cerebral cortex. Most patients with amoebic brain abscess have coexistent amoebic liver abscess, and are characterized by meningoencephalitis



Amoebic brain (cerebral) abscess



4. Amoebic cutaneous abscess: it occurs by direct extension around the anus and characterized by discharge sinuses from amoebic abscesses.

Cutaneous amoebiasis occur when trophozoites of *Entamoeba histolytica* site in the peripheral (perianal) of the anus, this lesion may be mistaken for candyloma or epithelioma.

This is variably the result of damaged skin frequently in the perianal region secondary to amoebic dysentery (intestinal amoebiasis). It is a venereal transmission among homosexual males (oral-anal).



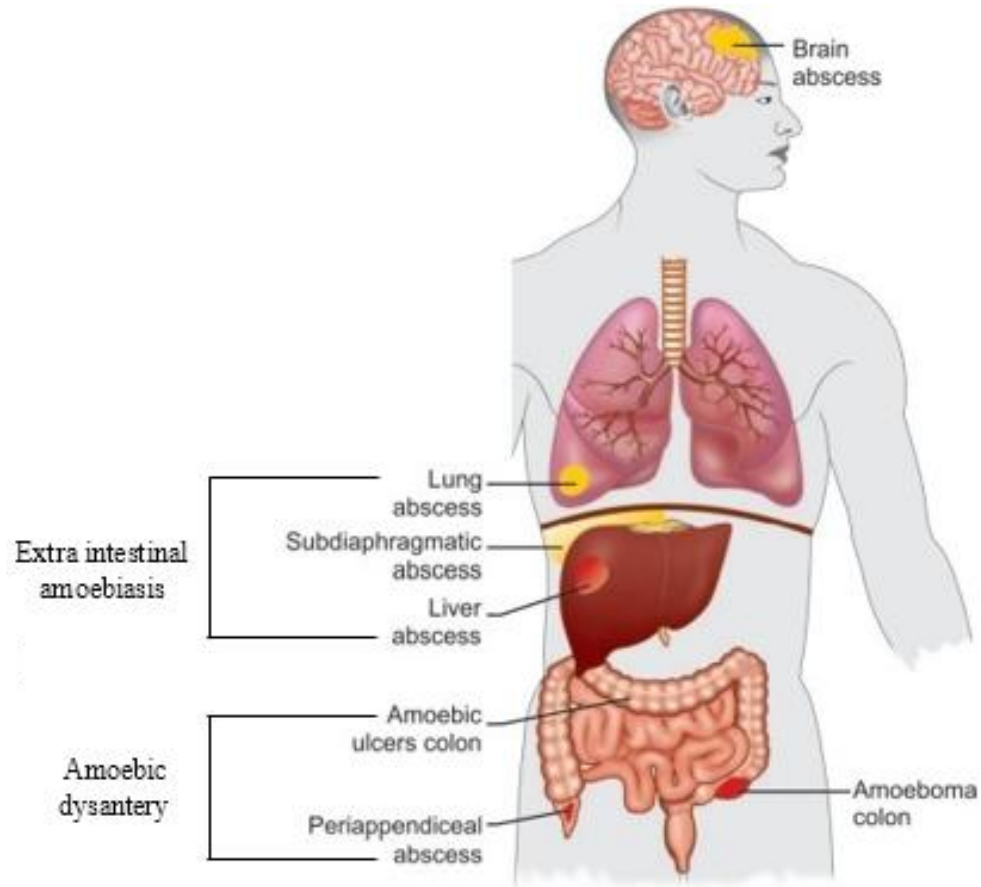
Amoebic cutaneous (perianal) abscess



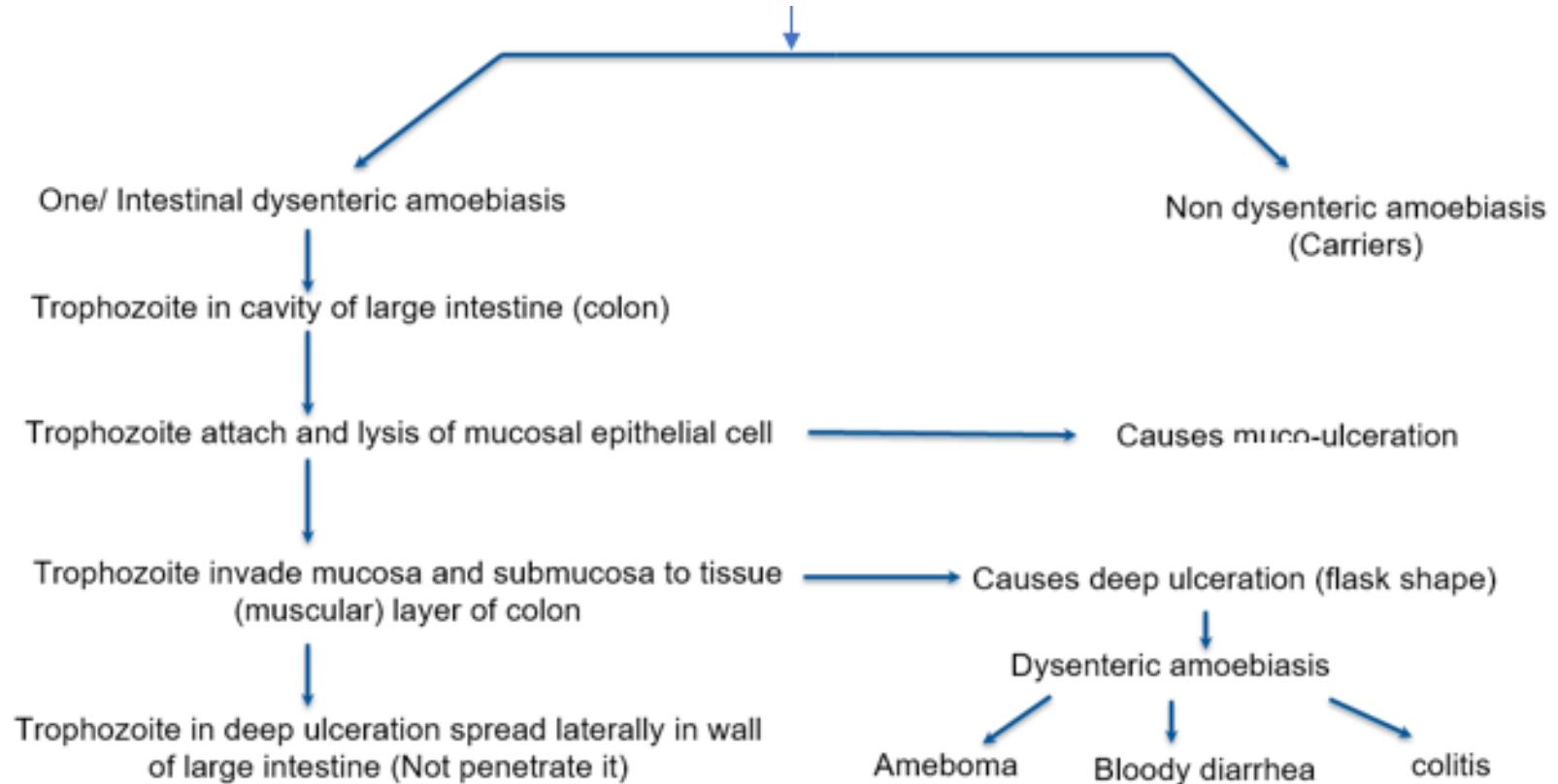
5. Amoebic genitourinary abscess: or renal amoebiasis is a rare complication to amoebic liver abscess, the prepuce and glans are affected in penile amoebiasis which is acquired through anal intercourse, similar lesions in females may occur on vulva, vagina, or cervix by spread from perineum.

The destructive ulcerative lesions resemble carcinoma.

Lesions of amoebiasis



Schematic diagram of disease causes by *Entamoeba histolytica* in large intestine and other organs of human body with the most important clinical symptoms



Extra intestinal amoebiasis (metastatic amoebiasis)

Trophozoite carried with blood in submucosa and tissue in wall in large intestine (Colon)

Trophozoite invade portal circulation → Causes amoebic liver abscess
and transform to liver

Trophozoite reach to lung → Causes amoebic lung (pulmonary) abscess

Trophozoite reach to Brain → Causes amoebic Brain (cerebral) abscess

Trophozoite reach to spleen → Causes amoebic splenic abscess

Trophozoite reach to Genitourinary system → Causes amoebic genitourinary abscess

Trophozoite reach (transform) to Perianal skin around anus → Causes amoebic skin(cutaneous) abscess



Laboratory diagnosis

I. Intestinal amoebiasis:

A. Macroscopic examination: The stool appearance is semi-liquid, foul smelling, brownish black in color, and intermingled with blood and mucus it does not adhere to the surface of the container.

B. Microscopic examination: Microscopic examination to patient stool shows trophozoite stage, cyst stage with RBC, and pus cells.



C. Stool culture: It is sensitive method to diagnosis of chronic and asymptomatic (carriers) of intestinal amoebiasis. There are different media to diagnosis of intestinal amoebiasis like **Diamonds medium** (casein hydrolysate) for culture and growth of *E. histolytica*, and Doeck & Drbohlav medium

D. Serological test: Indirect hemagglutination test (IHA), Elisa test (Enzyme-linked immunosorbent assay) and Latex agglutination test.

E. Molecular diagnosis: By detection of *E. histolytica* specific on DNA or antigen in stool samples by Probes Polymerase chain reaction (PCR) are more sensitive than microscopy it is a rapid and more specific method nowadays.



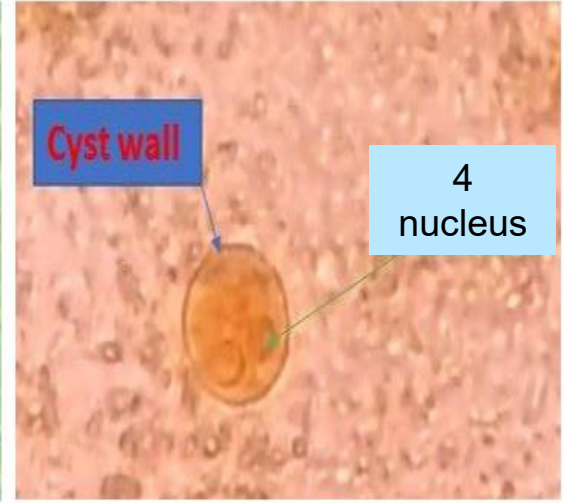
Extra intestinal amoebiasis: by serological test are the most important laboratory diagnosis to extra intestinal amoebiasis by complement fixation test, and indirect haemagglutination test (IHAT)



A



B



C

Microscopic examination of *Entamoeba histolytica* in patient stool

A. Trophozoite stage, B&C. Cyst stage

Treatment

Intestinal amoebiasis:

1. When stool contains cyst stage (carriers): only the treatment includes one of the antiamoebic are treated by **furamide (diloxamide furoate)** 500 mg orally three times daily for ten days, and **metronidazole (Flagyl)** is the drug of choice 800 mg orally every 8 hours for 5 days or 2 grams should be given orally once daily for three days.

2. When the stool contains trophozoite stage (dysenteric amoebiasis): Treated by **dehydroemetine** 1-1.5 mg/kg of body weight, intramuscular (IM) once daily for ten days.



Extra-intestinal amoebiasis: treated by combination of **dehydroemetine + Metronidazole (Flagyl)** or **furamide** or **chloraquine** (300 mg every 12 hour for two days). It's preferred to give antibiotics like **oxytetracycline** to prevent secondary bacterial infection



Prophylaxis

General prophylaxis is as for all fecal oral infections, food, water, and hand fingers have to be protected from contamination with humans, and not eating fresh uncooked vegetables or drinking unsterilized water and treated of carriers.



Thank you for listening, any questions?